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import { useState } from "react";

// — Full rec universe from the image —————
// Layer: D3S (source/root) | FO | MO | BO
// product: CEQ | LnD | MTN | OTC | SED | SEF | EQD | ETD | FX | ALL

const RECS = {
  // == D3S SOURCE RECS (4 PoC recs - break generators) =====
  D3S_CEQ: {
    id:"D3S_CEQ", layer:"D3S", product:"CEQ", office:"FO",
    label:"D3S Graph - CEQ Breaks",
    sublabel:"D3S OTC · Cash Equity breaks",
    desc:"Lavastorm D3S workflow generates CEQ reconciliation breaks. Root trigger",
    poc:true,
  },
  D3S_LND: {
    id:"D3S_LND", layer:"D3S", product:"LnD", office:"FO",
    label:"D3S Graph - LnD Breaks",
    sublabel:"D3S OTC · Loans & Deposits breaks",
    desc:"D3S workflow generates LnD breaks. Root trigger for downstream L&D cash",
    poc:true,
  },
  D3S_MTN: {
    id:"D3S_MTN", layer:"D3S", product:"MTN", office:"FO",
    label:"D3S Graph - MTN Breaks",
    sublabel:"D3S OTC · Medium Term Notes breaks",
    desc:"D3S workflow generates MTN breaks. Root trigger for coupon/maturity-dri",
    poc:true,
  },
  D3S_OTC: {
    id:"D3S_OTC", layer:"D3S", product:"OTC", office:"FO",
    label:"D3S Graph - OTC Breaks",
    sublabel:"D3S OTC · Derivatives breaks",
    desc:"D3S workflow generates OTC breaks covering EQD, ETD and FX derivatives.",
    poc:true,
  },

  // == FRONT OFFICE recs =====
  FOBO_EQ: {
    id:"FOBO_EQ", layer:"FO", product:"CEQ", office:"FO",
    label:"FOBO Rec - Equities",
    sublabel:"Trade population / position integrity",
    desc:"FO position rec between Front Office booking and Back Office confirmati",
  },

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FOBO_DIV: {
  id:"FOBO_DIV", layer:"FO", product:"CEQ", office:"FO",
  label:"FOBO Rec - Dividend",
  sublabel:"Corporate action event rec",
  desc:"Dividend event reconciliation. Timing breaks propagate to FLARE vs Fisc
},
FID_FTP: {
  id:"FID_FTP", layer:"FO", product:"OTC", office:"FO",
  label:"Fidessa vs FTP - Stock/Index Futures",
  sublabel:"ETD position population",
  desc:"ETD position rec between Fidessa and FTP. Root cause for FLARE vs HUB E
},

// == MIDDLE OFFICE recs =====
PL_CEQ: {
  id:"PL_CEQ", layer:"MO", product:"CEQ", office:"MO",
  label:"P&L Rec - CEQ",
  sublabel:"ACT / Product Control",
  desc:"Core CEQ P&L rec. Inherits from FOBO Equities and Sophis vs FTP Cash Eq
},
PL_SED: {
  id:"PL_SED", layer:"MO", product:"SED", office:"MO",
  label:"P&L Rec - SED",
  sublabel:"ACT / Product Control",
  desc:"SED P&L rec. Fed by FOBO Equities and Sophis FTP. Linked to CEQ through
},
PL_SEF: {
  id:"PL_SEF", layer:"MO", product:"SEF", office:"MO",
  label:"P&L Rec - SEF",
  sublabel:"ACT / Product Control",
  desc:"SEF P&L rec. Receives cascade from FOBO and FLARE breaks. Part of CEQ/S
},
SOPHIS_CEQ: {
  id:"SOPHIS_CEQ", layer:"MO", product:"CEQ", office:"MO",
  label:"Sophis vs FTP - Cash Equity (P&L - MTM & Cash)",
  sublabel:"Risk system / FTP valuation",
  desc:"MTM and cash P&L between Sophis and FTP. Receives CEQ position breaks f
},
SOPHIS_EQD: {
  id:"SOPHIS_EQD", layer:"MO", product:"OTC", office:"MO",
  label:"Sophis vs FTP - EQD (P&L - MTM & Cash)",
  sublabel:"Equity derivatives valuation",
  desc:"EQD MTM and cash P&L. Directly driven by D3S_OTC and Fidessa vs FTP ETD
},
SOPHIS_ETD: {
  id:"SOPHIS_ETD", layer:"MO", product:"OTC", office:"MO",
  label:"Sophis vs FTP - ETD PnL",

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    sublabel:"Exchange traded derivatives",
    desc:"ETD daily P&L. Driven by Fidessa vs FTP and D3S_OTC breaks. High false
},
SOPHIS_FX: {
    id:"SOPHIS_FX", layer:"MO", product:"FX", office:"MO",
    label:"Sophis vs FTP - FX Spot/Fwd/NDF/FX swap PnL",
    sublabel:"FX derivatives P&L",
    desc:"FX P&L rec across Spot, Forward, NDF and FX Swap. Linked to D3S_OTC for
},
SOPHIS_LND: {
    id:"SOPHIS_LND", layer:"MO", product:"LnD", office:"MO",
    label:"Sophis vs FTP - L&D (P&L - MTM & Cash)",
    sublabel:"Loans & Deposits valuation",
    desc:"L&D MTM and cash P&L. Direct downstream of D3S_LND breaks. Dedicated Ln
},
SOPHIS_MTN: {
    id:"SOPHIS_MTN", layer:"MO", product:"MTN", office:"MO",
    label:"Sophis vs FTP - MTN (P&L - MTM & Cash)",
    sublabel:"Medium Term Notes valuation",
    desc:"MTN MTM and cash P&L. Direct downstream of D3S_MTN breaks. Coupon and m
},
SOPHIS_VAULT: {
    id:"SOPHIS_VAULT", layer:"MO", product:"FX", office:"MO",
    label:"Sophis vs Vault (Murex) - FXO book level",
    sublabel:"FX Options / Murex",
    desc:"FX Options book level rec between Sophis and Murex Vault. Indirectly li
},
GLOBAL1: {
    id:"GLOBAL1", layer:"MO", product:"LnD", office:"MO",
    label:"Global1 vs FTP (HLR - SBL/FCT)",
    sublabel:"HLR funding / Prime Services",
    desc:"Funding and prime services rec. Receives cascade from D3S_LND and FLARE
},
HLR_PS: {
    id:"HLR_PS", layer:"MO", product:"LnD", office:"MO",
    label:"HLR - Prime Services",
    sublabel:"Prime brokerage / HLR",
    desc:"Prime services HLR rec. Linked to Global1 vs FTP and D3S_LND as indirec
},

// == BACK OFFICE recs ==
FLARE_FISCAL: {
    id:"FLARE_FISCAL", layer:"BO", product:"CEQ", office:"BO",
    label:"FLARE vs Fiscal Cash Control (Cash Equities)",
    sublabel:"Settlement / Fiscal accounting",
    desc:"Cash settlement rec between FLARE and Fiscal. Receives CEQ position bre
},

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FLARE_HUB_EQD: {
  id:"FLARE_HUB_EQD", layer:"BO", product:"OTC", office:"BO",
  label:"FLARE vs HUB Cash Contra (EQD Contra/control)",
  sublabel:"EQD cash contra / settlements",
  desc:"EQD cash contra between FLARE and HUB. Driven by D3S_OTC and Fidessa vs
},
FLARE_HUB_LND: {
  id:"FLARE_HUB_LND", layer:"BO", product:"LnD", office:"BO",
  label:"FLARE vs HUB Cash Contra (L&D)",
  sublabel:"L&D cash contra / settlements",
  desc:"L&D cash contra between FLARE and HUB. Direct downstream of D3S_LND bre
},
FLARE_SYN: {
  id:"FLARE_SYN", layer:"BO", product:"CEQ", office:"BO",
  label:"FLARE vs SYN - Cash Equities/SBL Control",
  sublabel:"Cash equities / SBL settlements",
  desc:"Cash equities and SBL control between FLARE and Synthesis. Driven by FO
},
NETTED: {
  id:"NETTED", layer:"BO", product:"CEQ", office:"BO",
  label:"Netted Cash Control - Cash Equities/SBL",
  sublabel:"Netting engine / settlements",
  desc:"Netted cash positions for CEQ and SBL. Receives residuals from FLARE vs
},
FID_FISCAL_FTP: {
  id:"FID_FISCAL_FTP", layer:"BO", product:"CEQ", office:"BO",
  label:"Fidessa vs Fiscal vs FTP - Cash Equity",
  sublabel:"Three-way cash equity rec",
  desc:"Three-way rec spanning Fidessa, Fiscal and FTP. Driven by D3S_CEQ and F
},
COMPTA_SED: {
  id:"COMPTA_SED", layer:"BO", product:"SED", office:"BO",
  label:"CRI COMPTA Rec - SED",
  sublabel:"COMPTA accounting / SED",
  desc:"SED COMPTA accounting rec. Receives cascade from FOBO breaks and D3S_CE
},
COMPTA_SEF: {
  id:"COMPTA_SEF", layer:"BO", product:"SEF", office:"BO",
  label:"CRI COMPTA Rec - SEF",
  sublabel:"COMPTA accounting / SEF",
  desc:"SEF COMPTA accounting rec. Linked to FOBO and D3S_CEQ cascade via SEF b
},
COMPTA_CEQ: {
  id:"COMPTA_CEQ", layer:"BO", product:"CEQ", office:"BO",
  label:"CRI COMPTA Rec - Cash Equities/SBL",
  sublabel:"COMPTA accounting / CEQ",
  desc:"Cash Equities COMPTA rec. Direct downstream of FLARE vs Fiscal and D3S_

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},
WASHBOOK: {
  id:"WASHBOOK", layer:"BO", product:"ALL", office:"BO",
  label:"Washbook '0006 Fiscal account' - Market Charges",
  sublabel:"Fiscal washbook / market charges",
  desc:"Market charges washbook. Receives indirect cascade from all four D3S re
},
};

// — Directed edges: from → to, with propagation mechanism —————
const EDGES = [
  // D3S_CEQ propagation chain
  { from:"D3S_CEQ", to:"FOBO_EQ",      prob:"H", mech:"CEQ position break surfac
  { from:"D3S_CEQ", to:"FLARE_FISCAL",  prob:"H", mech:"CEQ cash not posted to FL
  { from:"D3S_CEQ", to:"FID_FISCAL_FTP",prob:"H", mech:"CEQ position missing from
  { from:"D3S_CEQ", to:"SOPHIS_CEQ",    prob:"H", mech:"CEQ position difference →
  { from:"D3S_CEQ", to:"FLARE_SYN",     prob:"M", mech:"CEQ break propagates to F
  { from:"D3S_CEQ", to:"PL_CEQ",       prob:"H", mech:"CEQ position break → P&L

  // D3S_LND propagation chain
  { from:"D3S_LND", to:"FLARE_HUB_LND", prob:"H", mech:"LnD cash difference → FLA
  { from:"D3S_LND", to:"SOPHIS_LND",    prob:"H", mech:"LnD position break → Soph
  { from:"D3S_LND", to:"GLOBAL1",       prob:"M", mech:"LnD funding difference ca
  { from:"D3S_LND", to:"HLR_PS",        prob:"L", mech:"LnD break flows into prim
  { from:"D3S_LND", to:"WASHBOOK",     prob:"L", mech:"LnD market charges artefa

  // D3S_MTN propagation chain
  { from:"D3S_MTN", to:"SOPHIS_MTN",    prob:"H", mech:"MTN position break → Soph
  { from:"D3S_MTN", to:"WASHBOOK",     prob:"M", mech:"MTN coupon/market charges
  { from:"D3S_MTN", to:"FLARE_FISCAL",  prob:"M", mech:"MTN cash settlement feeds

  // D3S_OTC propagation chain
  { from:"D3S_OTC", to:"FID_FTP",       prob:"H", mech:"OTC/ETD position break su
  { from:"D3S_OTC", to:"FLARE_HUB_EQD", prob:"H", mech:"OTC EQD cash contra break
  { from:"D3S_OTC", to:"SOPHIS_EQD",    prob:"H", mech:"OTC EQD position → EQD MT
  { from:"D3S_OTC", to:"SOPHIS_ETD",    prob:"H", mech:"OTC ETD position → ETD Pn
  { from:"D3S_OTC", to:"SOPHIS_FX",     prob:"M", mech:"OTC FX component breaks →
  { from:"D3S_OTC", to:"SOPHIS_VAULT",  prob:"M", mech:"OTC FXO component → Sophi

  // Secondary cascades (FO → downstream)
  { from:"FOBO_EQ",      to:"FLARE_FISCAL",  prob:"H", mech:"Unmatched equity tr
  { from:"FOBO_EQ",      to:"NETTED",        prob:"H", mech:"FOBO position misma
  { from:"FOBO_EQ",      to:"PL_CEQ",        prob:"H", mech:"FOBO position break
  { from:"FOBO_EQ",      to:"PL_SED",        prob:"M", mech:"FOBO break shared a
  { from:"FOBO_EQ",      to:"PL_SEF",        prob:"M", mech:"FOBO break shared a
  { from:"FOBO_DIV",     to:"FLARE_FISCAL",  prob:"H", mech:"Dividend cash not p
  { from:"FOBO_DIV",     to:"PL_CEQ",        prob:"H", mech:"Dividend not booked

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    { from:"FOBO_DIV",      to:"COMPTA_CEQ",      prob:"M", mech:"Dividend entry miss
    { from:"FID_FTP",      to:"FLARE_HUB_EQD",    prob:"H", mech:"ETD position → EQD
    { from:"FID_FTP",      to:"SOPHIS_EQD",      prob:"H", mech:"ETD position miss →
    { from:"FID_FTP",      to:"SOPHIS_ETD",      prob:"H", mech:"Same position – ETD

// BO → BO secondary
    { from:"FLARE_FISCAL", to:"NETTED",          prob:"H", mech:"Unsettled cash → ne
    { from:"FLARE_FISCAL", to:"COMPTA_CEQ",      prob:"M", mech:"Missing settlement
    { from:"FLARE_SYN",    to:"NETTED",          prob:"M", mech:"SYN discrepancy flo

// MO → MO secondary
    { from:"SOPHIS_CEQ",   to:"PL_CEQ",          prob:"H", mech:"MTM difference is a
    { from:"SOPHIS_EQD",   to:"PL_CEQ",          prob:"M", mech:"EQD MTM feeds into
    { from:"GLOBAL1",      to:"HLR_PS",          prob:"M", mech:"HLR funding break p
];

// — Colour config —————
const LAYER = {
  D3S: { bg:"#0a0f1e", border:"#f59e0b", text:"#fde68a", badge:"#92400e", badgeTe
  FO:  { bg:"#0c1a2e", border:"#0ea5e9", text:"#bae6fd", badge:"#0c4a6e", badgeTe
  MO:  { bg:"#1a0f2e", border:"#a78bfa", text:"#ddd6fe", badge:"#2e1065", badgeTe
  BO:  { bg:"#0a1f14", border:"#10b981", text:"#a7f3d0", badge:"#064e3b", badgeTe
};

const PROD_COL = {
  CEQ:"#3b82f6", LnD:"#10b981", MTN:"#f59e0b",
  OTC:"#8b5cf6", SED:"#06b6d4", SEF:"#ec4899",
  EQD:"#0891b2", ETD:"#7c3aed", FX:"#d97706", ALL:"#64748b",
};

const PROB_COL = { H:"#ef4444", M:"#f59e0b", L:"#10b981" };
const PROB_LABEL = { H:"HIGH", M:"MED", L:"LOW" };

const BG = "#080d1a";

// — Tab config —————
const TABS = [
  { id:"cascade", label:"Cascade Map" },
  { id:"poc",      label:"PoC Proof" },
  { id:"matrix",  label:"Impact Matrix" },
];

// — Components —————
function ProdTag({ code }) {
  const c = PROD_COL[code] || "#64748b";
  return (
    <span style={{
      background: c + "22", border:`1px solid ${c}66`,
      color:c, borderRadius:3, padding:"1px 6px",

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        fontSize:9, fontWeight:700, letterSpacing:1,
    }}>{code}</span>
  );
}

function LayerTag({ layer }) {
  const l = LAYER[layer];
  return (
    <span style={{
      background:l.badge, color:l.badgeText,
      borderRadius:3, padding:"1px 7px", fontSize:9, fontWeight:700, letterSpacing:1,
    }}>{l.label}</span>
  );
}

function RecCard({ rec, highlight, dimmed, selected, onClick }) {
  const l = LAYER[rec.layer];
  const isPoc = rec.poc;
  return (
    <div onClick={() => onClick(rec.id)}
      style={{
        background: selected ? l.bg : (highlight ? l.bg+"cc" : "#0f1525"),
        border:`${selected ? 2.5 : 1.5}px solid ${selected ? l.border : (highlight ? l.border : "#f59e0b")}`,
        borderRadius:8, padding:"9px 11px", cursor:"pointer",
        opacity: dimmed ? 0.28 : 1,
        transition:"all 0.15s",
        boxShadow: isPoc ? `0 0 18px ${l.border}44` : selected ? `0 0 10px ${l.border}44`,
        position:"relative", marginBottom:6,
      }}>
      {isPoc && (
        <div style={{
          position:"absolute", top:-7, right:8,
          background:"#f59e0b", color:"#000",
          fontSize:8, fontWeight:900, padding:"1px 7px", borderRadius:3, letterSpacing:1,
        }}>D3S PoC</div>
      )}
      <div style={{display:"flex", gap:6, alignItems:"center", marginBottom:5, flexWrap:"wrap"}}>
        <LayerTag layer={rec.layer} />
        <ProdTag code={rec.product} />
        <span style={{color:"#475569", fontSize:9, fontFamily:"monospace"}}>{rec.product}</span>
      </div>
      <div style={{color: selected ? l.text : "#e2e8f0", fontSize:11, fontWeight:700}}>
        <div style={{color:"#475569", fontSize:9, fontStyle:"italic"}}>{rec.sublabel}</div>
      </div>
    </div>
  );
}

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function EdgeCard({ edge, direction }) {
  const other = direction === "down" ? RECS[edge.to] : RECS[edge.from];
  const l = LAYER[other.layer];
  const pc = PROB_COL[edge.prob];
  return (
    <div style={{
      background:"#0a0e1a", border:`1px solid ${l.border}44`,
      borderLeft:`4px solid ${l.border}`,
      borderRadius:"0 7px 7px 0", padding:"9px 13px", marginBottom:8,
    }}>
      <div style={{display:"flex", gap:7, alignItems:"center", marginBottom:5, fl
        <span style={{color:l.border, fontSize:14, fontWeight:900}}>{direction===
        <LayerTag layer={other.layer} />
        <ProdTag code={other.product} />
        <span style={{
          marginLeft:"auto", background:pc+"22", border:`1px solid ${pc}66`,
          color:pc, borderRadius:3, padding:"1px 6px", fontSize:8, fontWeight:800
        }}>{PROB_LABEL[edge.prob]} PROB</span>
      </div>
      <div style={{color:l.text, fontWeight:700, fontSize:11, marginBottom:3}}>{o
        <div style={{color:"#64748b", fontSize:10, lineHeight:1.5}}>{edge.mech}</di
      </div>
    );
  }

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// — Tab: Cascade Map —————
function CascadeTab() {
  const [selected, setSelected] = useState(null);
  const [layerFilter, setLayerFilter] = useState("ALL");

  const toggle = (id) => setSelected(s => s === id ? null : id);

  const downEdges = selected ? EDGES.filter(e => e.from === selected) : [];
  const upEdges    = selected ? EDGES.filter(e => e.to    === selected) : [];
  const touched    = selected ? new Set([...downEdges.map(e=>e.to), ...upEdges.map

  const allRecs = Object.values(RECS);
  const filtered = layerFilter === "ALL" ? allRecs : allRecs.filter(r => r.layer
  const byLayer = { D3S:[], FO:[], MO:[], BO:[] };
  filtered.forEach(r => byLayer[r.layer]?.push(r));

  const selRec = selected ? RECS[selected] : null;

  return (
    <div style={{display:"flex", gap:14, minHeight:500}}>
      {/* Left: rec list */}
      <div style={{width:260, flexShrink:0}}>

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    { /* Layer filter */ }
    <div style={{display:"flex", gap:5, marginBottom:12, flexWrap:"wrap"}}>
      {[ "ALL", "D3S", "FO", "MO", "BO" ].map(l => (
        <button key={l} onClick={()=>setLayerFilter(l)} style={{
          background: layerFilter===l ? (LAYER[l]?.badge||"#334155") : "none"
          border:`1px solid ${layerFilter===l ? (LAYER[l]?.border||"#64748b")
          color: layerFilter===l ? "#fff" : "#64748b",
          borderRadius:4, padding:"3px 9px", cursor:"pointer",
          fontFamily:"inherit", fontSize:9, fontWeight:700, letterSpacing:1,
        }}>{l}</button>
      ) ) }
    </div>
    {Object.entries(byLayer).map(([layer, recs]) => recs.length === 0 ? null
      <div key={layer} style={{marginBottom:14}}>
        <div style={{
          color:LAYER[layer].border, fontSize:9, fontWeight:800,
          letterSpacing:2, textTransform:"uppercase", marginBottom:6,
          borderBottom:`1px solid ${LAYER[layer].border}44`, paddingBottom:3,
        }}>{LAYER[layer].label} ({recs.length})</div>
        {recs.map(r => (
          <RecCard key={r.id} rec={r}
            selected={selected === r.id}
            highlight={touched?.has(r.id) && selected !== r.id}
            dimmed={touched && !touched.has(r.id)}
            onClick={toggle} />
        ) ) }
      </div>
    ) ) }
  </div>

  { /* Right: detail */ }
  <div style={{flex:1}}>
    { !selected ? (
      <div style={{
        background:"#0f1525", border:"1px solid #1e2d4a",
        borderRadius:10, padding:"32px 24px", textAlign:"center",
      }}>
        <div style={{fontSize:36, marginBottom:12}}>🔗</div>
        <div style={{color:"#e2e8f0", fontWeight:700, fontSize:15, marginBott
        <div style={{color:"#475569", fontSize:11, lineHeight:1.7, marginBott
          Click a rec to trace its full upstream root cause chain<br/>and all
        </div>
        <div style={{color:"#64748b", fontSize:10, fontWeight:700, marginBott
        { [
          {d:"D3S_CEQ", chain:"→ FOBO Equities → FLARE Fiscal → Netted Cash →
          {d:"D3S_LND", chain:"→ FLARE HUB L&D → Sophis L&D P&L → Global1 HLR
          {d:"D3S_MTN", chain:"→ Sophis MTN P&L → Washbook 0006 → FLARE Fisca

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    {d:"D3S_OTC", chain:"→ Fidessa FTP → FLARE HUB EQD → Sophis EQD/ETD
].map(item => (
  <div key={item.d} onClick={() => toggle(item.d)}
    style={{
      background:"#0a0e1a", border:`1px solid ${item.c}44`,
      borderLeft:`3px solid ${item.c}`, borderRadius:"0 6px 6px 0",
      padding:"7px 12px", textAlign:"left", marginBottom:6, cursor:"p
    }}>
    <span style={{color:item.c, fontWeight:800, fontSize:10}}>{item.d
    <span style={{color:"#64748b", fontSize:10}}>{item.chain}</span>
  </div>
  )))
</div>
) : (
<div>
  {/* Selected rec header */}
  <div style={{
    background:LAYER[selRec.layer].bg,
    border:`2px solid ${LAYER[selRec.layer].border}`,
    borderRadius:9, padding:"14px 18px", marginBottom:14,
  }}>
    <div style={{display:"flex", gap:8, alignItems:"center", marginBott
      <LayerTag layer={selRec.layer} />
      <ProdTag code={selRec.product} />
      {selRec.poc && <span style={{background:"#f59e0b", color:"#000",
    </div>
    <div style={{color:LAYER[selRec.layer].text, fontWeight:800, fontSi
      <div style={{color:"#64748b", fontSize:11, lineHeight:1.5}}>{selRec
    </div>
  {/* Stats */}
  <div style={{display:"flex", gap:10, marginBottom:14, flexWrap:"wrap"
    [[
      {n:upEdges.length, label:"upstream causes", c:"#ef4444"},
      {n:downEdges.length, label:"downstream impacts", c:"#3b82f6"},
      {n:downEdges.filter(e=>e.prob==="H").length, label:"high-prob lin
    ].map(s => (
      <div key={s.label} style={{background:"#0f1525", border:`1px soli
        <span style={{color:s.c, fontWeight:800, fontSize:18}}>{s.n} </
        <span style={{color:"#64748b", fontSize:10}}>{s.label}</span>
      </div>
    )))
  </div>
  {/* Upstream */}
  {upEdges.length > 0 && (
    <div style={{marginBottom:16}}>
      <div style={{color:"#ef4444", fontSize:9, fontWeight:800, letters
        ▲ UPSTREAM ROOT CAUSES – breaks in these recs drive this break

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        </div>
        {upEdges.map((e,i) => <EdgeCard key={i} edge={e} direction="up" /
    </div>
    )}
    {/* Downstream */}
    {downEdges.length > 0 && (
        <div>
            <div style={{color:"#3b82f6", fontSize:9, fontWeight:800, letters
                ▼ DOWNSTREAM PROPAGATION – this break will cause breaks in thes
            </div>
            {downEdges.map((e,i) => <EdgeCard key={i} edge={e} direction="dow
        </div>
    )}
    </div>
    )}
    </div>
    </div>
    </div>
    );
}

```

// — Tab: PoC Proof —————

```

function PocTab() {
    const [step, setStep] = useState(0);

    const STEPS = [
        {
            title:"Step 1 – D3S Generates a Break",
            poc:"D3S_CEQ", color:"#f59e0b",
            icon:"🔍",
            desc:"The Lavastorm D3S workflow for CASHFOARE runs and generates a CEQ bre
            evidence:"D3S .lod file parsed → structured JSON → break_id: CEQ-2024-001,
            mlAction:"false_break_prob scored at 0.82 → risk_tier: Suppress (amount < £
        },
        {
            title:"Step 2 – Break Propagates to FOBO Rec",
            poc:"FOBO_EQ", color:"#0ea5e9",
            icon:"→",
            desc:"The CEQ position break from D3S surfaces in FOBO Rec – Equities withi
            evidence:"Break detected: FOBO_EQ · entity: ENTITY_A · amount: £2,340,000 ·
            mlAction:"upstream_d3s_ceq_break = 1 (same entity, date, ~amount match). ML
        },
        {
            title:"Step 3 – Cascades to FLARE vs Fiscal",
            poc:"FLARE_FISCAL", color:"#10b981",
            icon:"→",
            desc:"The unmatched CEQ position in FOBO means the corresponding cash posti
            evidence:"Break detected: FLARE_FISCAL · entity: ENTITY_A · amount: £2,340,

```

```

    mlAction:"upstream_d3s_ceq_break = 1, upstream_fol_break = 1. ML scores FLA
  },
  {
    title:"Step 4 – Cascades to Netted Cash Control",
    poc:"NETTED", color:"#10b981",
    icon:"→",
    desc:"The FLARE vs Fiscal break means the settlement doesn't net correctly,
    evidence:"Break detected: NETTED · entity: ENTITY_A · amount residual: £2,3
    mlAction:"upstream_d3s_ceq_break = 1, upstream_flare_fiscal_break = 1. ML f
  },
  {
    title:"Step 5 – P&L Rec – CEQ Inherits",
    poc:"PL_CEQ", color:"#a78bfa",
    icon:"→",
    desc:"The CEQ position difference creates a P&L exposure in P&L Rec – CEQ.
    evidence:"Break detected: PL_CEQ · entity: ENTITY_A · P&L impact: £187,500
    mlAction:"upstream_d3s_ceq_break = 1, upstream_fol_break = 1, mad_score = 1
  },
  {
    title:"Step 6 – PoC Proven",
    poc:null, color:"#f59e0b",
    icon:"✓",
    desc:"One D3S_CEQ root cause break generates 5 downstream breaks across 5 d
    evidence:"Root cause: D3S_CEQ · Downstream breaks: FOBO_EQ, FLARE_FISCAL, N
    mlAction:"PoC demonstrates: cross-rec upstream flags + D3S break detection
  },
];

```

```

const s = STEPS[step];

```

```

return (
  <div>
    {/* Step progress */}
    <div style={{display:"flex", gap:6, marginBottom:20, flexWrap:"wrap"}}>
      {STEPS.map((st, i) => (
        <button key={i} onClick={() => setStep(i)} style={{
          background: step===i ? s.color+"33" : "none",
          border:`1px solid ${step===i ? s.color : "#1e2d4a"}`,
          color: step===i ? s.color : "#475569",
          borderRadius:5, padding:"5px 12px", cursor:"pointer",
          fontFamily:"inherit", fontSize:9, fontWeight:800,
        }}>Step {i+1}</button>
      )
      )}
    </div>

    {/* Step content */}
    <div style={{background:"#0f1525", border:`2px solid ${s.color}55`, borderR

```

```

<div style={{display:"flex", gap:14, alignItems:"flex-start", marginBotto
  <div style={{
    width:48, height:48, borderRadius:"50%",
    background:s.color+"22", border:`2px solid ${s.color}`,
    display:"flex", alignItems:"center", justifyContent:"center",
    fontSize:22, flexShrink:0,
  }}>{s.icon}</div>
<div>
  <div style={{color:s.color, fontSize:10, fontWeight:800, letterSpacing:
    STEP {step+1} OF {STEPS.length}
  </div>
  <div style={{color:"#f1f5f9", fontWeight:800, fontSize:16}}>{s.title}
</div>
  {s.poc && RECS[s.poc] && (
    <div style={{marginLeft:"auto"}}>
      <RecCard rec={RECS[s.poc]} selected={false} highlight={true} dimmed
    </div>
  )}
</div>

<div style={{color:"#94a3b8", fontSize:12, lineHeight:1.7, marginBottom:1

<div style={{background:"#070b15", border:`1px solid ${s.color}33`, borde
  <div style={{color:s.color, fontSize:9, fontWeight:800, letterSpacing:2
    <div style={{color:"#e2e8f0", fontSize:11, fontFamily:"monospace", line
  </div>

<div style={{background:"#0a1020", border:"1px solid #a78bfa44", borderLe
  <div style={{color:"#a78bfa", fontSize:9, fontWeight:800, letterSpacing
    <div style={{color:"#94a3b8", fontSize:11, lineHeight:1.6}}>{s.mlAction
  </div>

<div style={{display:"flex", gap:10, marginTop:16}}>
  {step > 0 && (
    <button onClick={()=>setStep(s=>s-1)} style={{
      background:"none", border:"1px solid #1e2d4a",
      color:"#64748b", borderRadius:5, padding:"7px 16px",
      cursor:"pointer", fontFamily:"inherit", fontSize:10,
    }}>← Previous</button>
  )}
  {step < STEPS.length-1 && (
    <button onClick={()=>setStep(s=>s+1)} style={{
      background:s.color+"22", border:`1px solid ${s.color}`,
      color:s.color, borderRadius:5, padding:"7px 16px",
      cursor:"pointer", fontFamily:"inherit", fontSize:10, fontWeight:700
      marginLeft:"auto",
    }}>Next Step →</button>
  )}

```

```

    })
  </div>
</div>

{/* Chain visualiser */}
<div style={{marginTop:16, background:"#070b15", border:"1px solid #1e2d4a"
  <div style={{color:"#64748b", fontSize:9, fontWeight:800, letterSpacing:2
    <div style={{display:"flex", alignItems:"center", gap:4, flexWrap:"wrap"}
      [{"D3S_CEQ","FOBO_EQ","FLARE_FISCAL","NETTED","PL_CEQ"].map((id, i) =>
        const r = RECS[id];
        const active = step > i - (id==="D3S_CEQ" ? 0 : 1);
        return (
          <div key={id} style={{display:"flex", alignItems:"center", gap:4}}>
            {i > 0 && <div style={{color: active ? "#f59e0b" : "#1e2d4a", fon
              <div style={{
                background: active ? LAYER[r.layer].bg : "#0f1525",
                border:`1.5px solid ${active ? LAYER[r.layer].border : "#1e2d4a
                borderRadius:6, padding:"5px 10px",
                opacity: active ? 1 : 0.35,
              }}>
                <div style={{color: active ? LAYER[r.layer].text : "#475569", f
                  <div style={{color:"#475569", fontSize:8}}>{r.product}</div>
                </div>
              </div>
            </div>
          </div>
        );
      )}}
    </div>
  </div>
  </div>
  </div>
);
}

```

```

// — Tab: Impact Matrix —————
function MatrixTab() {
  const D3S_IDS = ["D3S_CEQ","D3S_LND","D3S_MTN","D3S_OTC"];
  const allOther = Object.values(RECS).filter(r => !r.poc);

  // For each downstream rec, find which D3S recs drive it
  function getImpact(recId, d3sId) {
    // Direct edge
    if (EDGES.find(e => e.from === d3sId && e.to === recId)) {
      return EDGES.find(e => e.from === d3sId && e.to === recId);
    }
    // Indirect (2 hops)
    const hop1 = EDGES.filter(e => e.from === d3sId);
    for (const h of hop1) {
      const hop2 = EDGES.find(e => e.from === h.to && e.to === recId);
    }
  }
}

```

```

    if (hop2) return { ...hop2, indirect:true, prob: hop2.prob === "H" ? "M" :
    }
    return null;
}

const D3S_META = {
  D3S_CEQ:{ c:"#3b82f6", label:"D3S CEQ" },
  D3S_LND:{ c:"#10b981", label:"D3S LnD" },
  D3S_MTN:{ c:"#f59e0b", label:"D3S MTN" },
  D3S_OTC:{ c:"#8b5cf6", label:"D3S OTC" },
};

// Only show recs that have at least one impact
const impacted = allOther.filter(r => D3S_IDS.some(d => getImpact(r.id, d)));

return (
  <div style={{overflowX:"auto"}}>
    <div style={{color:"#64748b", fontSize:10, marginBottom:12}}>
      Which downstream recs are impacted by each D3S PoC break source. Solid =
    </div>
    <table style={{width:"100%", borderCollapse:"collapse", fontSize:10}}>
      <thead>
        <tr>
          <th style={{color:"#64748b", textAlign:"left", padding:"6px 10px", ba
            DOWNSTREAM REC
          </th>
          {D3S_IDS.map(d => (
            <th key={d} style={{padding:"6px 10px", background:"#0f1525", borde
              <div style={{color:D3S_META[d].c, fontWeight:800, fontSize:9, let
            </th>
          )))
          <th style={{color:"#64748b", textAlign:"center", padding:"6px 10px",
        </tr>
      </thead>
      <tbody>
        {impacted.map((rec, ri) => {
          const l = LAYER[rec.layer];
          return (
            <tr key={rec.id} style={{background: ri%2===0 ? "#080d1a" : "#0a0f1
              <td style={{padding:"6px 10px", borderBottom:"1px solid #111827"}
                <div style={{color:"#e2e8f0", fontWeight:700, fontSize:10}}>{re
                  <div style={{color:"#475569", fontSize:8}}>{rec.id}</div>
                </td>
                {D3S_IDS.map(d => {
                  const imp = getImpact(rec.id, d);
                  const pc = imp ? PROB_COL[imp.prob] : null;
                  return (

```

```

        <td key={d} style={{padding:"5px 8px", textAlign:"center", bo
        {imp ? (
            <div style={{
                background: imp.indirect ? "none" : pc+"33",
                border:`${imp.indirect ? "1.5px dashed" : "1px solid"}
                borderRadius:4, padding:"3px 5px", display:"inline-bloc
            }}>
                <div style={{color:pc, fontWeight:800, fontSize:8}}>{PR
                <div style={{color:"#475569", fontSize:7}}>{imp.indirec
            </div>
        ) : (
            <span style={{color:"#1e2d4a", fontSize:12}}>—</span>
        )}
    </td>
    </tr>
    </tbody>
    </table>
    </div>
    );
}

```

```

// — App —————
export default function App() {
    const [tab, setTab] = useState("cascade");

    const pocCount    = Object.values(RECS).filter(r=>r.poc).length;
    const totalEdges  = EDGES.length;
    const highEdges    = EDGES.filter(e=>e.prob==="H").length;
    const recCount     = Object.keys(RECS).length;

    return (
        <div style={{
            background:BG, minHeight:"100vh",
            fontFamily:"'IBM Plex Mono','Courier New',monospace",
            color:"#e2e8f0", padding:"20px 16px",
        }}>
            { /* Header */}

```



```

<div style={{marginBottom:20}}>
  <div style={{
    display:"inline-block", background:"#f59e0b22",
    border:"1px solid #f59e0b55", borderRadius:5,
    padding:"3px 12px", color:"#f59e0b",
    fontSize:9, fontWeight:800, letterSpacing:2, marginBottom:10,
  }}>CASHFOARE · D3S PoC · ROOT CAUSE CASCADE MAP</div>
  <h1 style={{margin:"0 0 5px", fontSize:"clamp(16px,3vw,22px)", fontWeight
    Upstream → Downstream Break Linkage
  </h1>
  <p style={{color:"#475569", fontSize:11, margin:0}}>
    4 D3S graphs as root cause sources · 27 recs · full propagation chain m
  </p>
</div>

{/* KPI strip */}
<div style={{display:"flex", gap:10, marginBottom:20, flexWrap:"wrap"}}>
  [[
    {v:pocCount, l:"D3S PoC Sources", c:"#f59e0b"},
    {v:recCount, l:"Total Recs Mapped", c:"#0ea5e9"},
    {v:totalEdges, l:"Propagation Links", c:"#a78bfa"},
    {v:highEdges, l:"High-Prob Links", c:"#ef4444"},
  ].map(k => (
    <div key={k.l} style={{
      background:"#0f1525", border:`1px solid #1e2d4a`,
      borderTop:`3px solid ${k.c}`, borderRadius:7,
      padding:"8px 16px", flex:1, minWidth:90,
    }}>
      <div style={{color:k.c, fontWeight:900, fontSize:22}}>{k.v}</div>
      <div style={{color:"#475569", fontSize:9}}>{k.l}</div>
    </div>
  )))
</div>

{/* Tabs */}
<div style={{display:"flex", gap:3, marginBottom:20, borderBottom:"1px soli
  {TABS.map(t => (
    <button key={t.id} onClick={()=>setTab(t.id)} style={{
      background:"none", border:"none",
      borderBottom: tab===t.id ? "2px solid #f59e0b" : "2px solid transpare
      color: tab===t.id ? "#f59e0b" : "#475569",
      fontFamily:"inherit", fontSize:10, fontWeight:800,
      padding:"8px 16px", cursor:"pointer", letterSpacing:1, textTransform:
    }}>{t.label}</button>
  )))
</div>

```

```
{tab === "cascade" && <CascadeTab />}
{tab === "poc"      && <PocTab />}
{tab === "matrix"   && <MatrixTab />}

<div style={{color:"#1e2d4a", fontSize:9, textAlign:"center", marginTop:28,
  CASHFOARE · ACT · OMRC AI/ML · D3S PoC Root Cause Engine · April
</div>
</div>
);
}
```